Supporting calculations for The value of prior information for final dose selection in BOIN trials

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This note gives the additional model specifications, comparisons and computational records for the combined article. The article contains the method and its proofs. The accompanying data define the evaluated rules. Fresh searches can return different feasible rules because optimization is time limited.

1 Planning information and available gain

The six-dose study uses target $\varphi = .25$, cohorts of three and at most 30 participants, starting at dose 1. Its planning support contains 144 curves with positive weight and additional guard curves with zero planning weight. Location weights are

$$q^{\text{central}} = (.02, .04, .42, .42, .08, .02), \qquad q^{\text{high}} = (.01, .02, .04, .08, .20, .65).$$

The third specification weights the six positions equally. Within each position, the nominal conditional distribution preserves the supplied curve weights.

Let H_{jf} denote the conditional distribution for shape family f at MTD position j. Only

families represented with positive weight at that position enter the class. The support has 55 such groups. For a fixed rule T, define

$$z_{jf}(T) = E_{H_{jf}}\{P_p(T = j) - P_p(b = j)\}.$$

Allowing a separate mixture of these families at every position gives the smallest gain

$$\min_s \sum_j q_j^s \min_f z_{jf}(T).$$

The high-location analysis uses one location specification. The joint analysis uses the convex hull of central, high and equal location weights. These classes preserve the represented distributions within each family. They do not include arbitrary new dose–toxicity shapes.

Construction introduces one auxiliary value for each position, bounded above by every corresponding family gain. A further auxiliary value is bounded by each weighted location sum. Maximizing the latter value gives the stated worst-family objective. Risk constraints and eligibility are the same as in the nominal study. Endpoint recommendations and one reference outcome per dose remain fixed.

For the upper bound on attainable gain, a nonnegative mixture of the location vertices and normalized family weights gives an explicit member of the planning class. A bound on gain under that member also bounds the smallest gain over the class. The independent calculation checks this membership, the permitted choices and all nonnegative risk prices before applying the weak-duality inequality in the article. The resulting bounds cover deterministic and randomized rules under the stated restrictions.

Table 1: Worst-family planning gain and upper bounds, percentage points. Construction upper is the numerical search bound. Overall upper uses independent likelihood enclosures and exact rational arithmetic. Upper bounds are rounded upward.

Planning class	Attained	Construction upper	Overall upper
High locations, varying shapes	.513	.520	.891
Varying locations and shapes	.116	.132	.261

The supplied implementation encloses terminal probabilities with outward-rounded integer arithmetic. Prior weights are interpreted as their saved binary values and normalized with exact rational arithmetic. Signed summation then gives lower and upper gain bounds. The risk and gain checks are independent of the numerical optimizer. The construction upper bounds are numerical optimization results.

2 Bayesian dose–response comparison

The logistic comparison adapts the posterior-mean final selection of Wakayama et al. (2025). Their study uses six doses, target .30, 36 participants, three link functions and Hamiltonian Monte Carlo. The present comparison uses target .25, 30 participants and the logit link on doses $(7, 14, 29, 51, 103, 188)$ mg. The model is

$$\operatorname{logit}(p_j) = \beta_0 + \exp(\beta_1)\log(d_j/51), \qquad \beta_0 \sim N(\mu_0, \sigma_0^2), \quad \beta_1 \sim N(\mu_1, \sigma_1^2),$$

with independent priors.

Endpoint beliefs are $P(p_1 > \varphi) = .05$ and $P(p_6 \leq \varphi) = .21$. Beta marginals express these beliefs. Their medians are interpolated on the logit scale across log doses. The model prior is fitted to the .025, .5 and .975 marginal toxicity quantiles. Calibration depends on the physical grid and the induced toxicity distributions (Neuenschwander et al., 2008; Alger et al., 2026). A broader prior fixes the log-slope standard deviation at .75 and refits the other

parameters. Table 2 summarizes the two fitted priors. The supplied inputs retain the full precision.

Table 2: Calibrated logistic priors. Standard deviations are on the intercept and log-slope scales.

Prior	μ_0	σ_0	μ_1	σ_1
Quantile	-1.171	1.320	.102	.010
Broader	-1.270	1.201	.050	.750

Gaussian quadrature at two integration orders checks the posterior mean toxicities and proposed actions. The calculation uses 181 and 241 nodes per dimension. A separate integral over log slope checks the fitted marginal quantiles. The posterior likelihood uses the complete terminal counts. Path coefficients cancel in posterior normalization.

At each terminal outcome, the proposal selects the eligible dose whose posterior mean toxicity is closest to target. Endpoint choices, reference outcomes and no-recommendation status are then retained. The risk filter chooses either that proposal or BOIN. Central and high-location filters have their respective nominal planning objectives. Both logistic priors use the same grid, eligibility, risk limits and six requested eight-second search rounds as the finite-prior comparisons.

The three independent evaluation libraries each contain 576 curves: 24 in each of four shape families at every MTD position. Rules are fixed before these libraries are generated. Table D.1 of the combined article reports mean PCS and mean gain across the libraries, with the smallest curve-level gain across all three. These are comparisons of the specified adaptations under common risk requirements.

3 Uncertainty from generating evaluation curves

Terminal-outcome summation gives exact operating characteristics for a fixed toxicity curve, up to numerical evaluation. The primary average also varies because its curves are sampled. Each of its four generator families supplies 40 independent curves at each of six MTD positions. For paired differences d_{jfi},

$$\widehat{\Delta} = \sum_{jf} (q_j/4)\bar{d}_{jf}, \qquad \mathrm{SE}(\widehat{\Delta}) = \left\{ \sum_{jf} (q_j/4)^2 s_{jf}^2/n_{jf} \right\}^{1/2}.$$

Table 3 uses the normal approximation $\widehat{\Delta} \pm 1.96\,\mathrm{SE}$. It describes sampling under these generators. It does not measure uncertainty about a future compound or about clinical adoption.

Table 3: Primary mean PCS gains and approximate 95% generator intervals, percentage points.

Prior	Rule	Mean gain	Interval
Central	Full table	.575	(.505, .646)
Central	80% table	.522	(.467, .577)
Central	90% table	.578	(.517, .638)
High	Full table	.522	(.491, .553)
High	80% table	.396	(.366, .427)
High	90% table	.461	(.430, .492)

4 Comparison of risk bounds

The ordered-likelihood and Continuous Simulation Extension (CSE) calculations start from the same 1,716 cells and target a two-point risk bound. They use the same fixed central and high rules. Interior PCS-loss and both signs of each tail contrast are checked. The CSE

implementation uses exact terminal expectations at each knowledge point (Yang et al., 2024).

Write L for the likelihood ratio from knowledge point p^0 to p. For any event A and $r > 1$, the tilt bound gives

$$P_p(A) \le P_{p^0}(A)^{1-1/r}\{E_{p^0}(L^r)\}^{1/r}.$$

Positive and negative events in a signed contrast are bounded separately. The negative-event lower bound uses both its complement and the reverse likelihood ratio. An alternative applies the tilt bound to the contrast shifted and scaled between zero and one. The calculation retains the smallest valid upper bound.

The independent-sequence embedding supplies each dose with its largest reachable enrollment. A second embedding accounts for bounded stopping. For a cell with coordinate bounds (ℓ_j, u_j), define

$$\bar{\psi}_j = \max_{t\in[\ell_j,u_j]} \log\{t^r(p_j^0)^{1-r} + (1-t)^r(1-p_j^0)^{1-r}\}.$$

The quantity before taking the logarithm is convex, so the coordinate maximum occurs at an endpoint. A nonnegative supermartingale and bounded stopping give

$$E_{p^0}(L_\tau^r) \le \exp\left\{\max_{n\in\mathcal{N}} \sum_j n_j \bar{\psi}_j\right\},$$

where $\mathcal{N}$ contains the reachable final enrollment vectors. The reverse calculation exchanges the two probability measures.

All methods bisect unresolved cells along their widest coordinate. Children retain the smaller of their new bound and their parent's bound. The original comparison gives each method 60 seconds after initialization and at most 200,000 leaves. Its timings and refinement histories are supplied. Replaying to the recorded final refinement counts checks the reported bounds without making results depend on the speed of a later computer. A separate command repeats the resource-limited benchmark on the current computer. Independent moment checks

compare the stopped-trial bound with direct terminal likelihood-ratio sums.

CSE results are numerical tilt calculations. The completed ordered-likelihood calculation also receives the independent integer risk check used for the final selection rules. Table 6 of the article reports the original comparison.

5 Other trial designs and computational records

The supporting designs vary maximum enrollment, grid size and cohort size. They use target .25 and starting dose 1. Their planning curves have the form

$$p_i = \text{logit}^{-1}\{\text{logit}(\varphi) + s(i - j + t)\}.$$

The six slopes are $.3, .5, .8, 1.2, 1.8, 2.6$. Interior offsets are $\pm.08$ and $\pm.22$. Endpoint offsets point inward with magnitudes $.04, .08, .15, .22$. The 24 curves at each position have equal conditional weight. Six-dose location weights match the main study. The four-dose weights are $(.04, .46, .46, .04)$ and $(.03, .08, .24, .65)$. The eight-dose weights are

$$q^{\text{central}} = (.02, .03, .10, .35, .35, .10, .03, .02),$$
$$q^{\text{high}} = (.01, .01, .02, .03, .05, .08, .15, .65).$$

Table 4: Supporting designs on fixed descriptive evaluation libraries. BOIN PCS is a percentage. Signed and Separate are changes in percentage points. Worst is the smallest curve-level change for the signed rule.

Design	Prior	BOIN PCS	Signed	Separate	Worst
6 doses, 36 participants	Central	35.159	.660	.545	-1.701
6 doses, 36 participants	High	29.877	.305	.294	-1.263
8 doses, 24 participants	Central	26.162	.363	.367	-1.498
8 doses, 24 participants	High	13.404	.055	.053	-1.370
4 doses, cohorts of 2	Central	36.547	-.004	.013	-1.047
4 doses, cohorts of 2	High	35.032	.697	.494	-1.622

Signed rules use both signs of each risk contrast within ordered cells. Separate rules bound adverse contributions state by state. The supporting calculations rebuild each catalog, check every saved rule, and recompute its operating characteristics and planning gains. The eight-dose partition starts from five coordinate intervals. The other supporting partitions start from eight. Each final partition has 1,716 cells. The broader search can retain a feasible comparator when the time-limited search does not find a better rule.

The original nominal workflow takes 833.3 seconds, including initialization, enumeration, prior integration, optimization, verification and decision-table export. Its largest child process uses 4,588 MiB. Given those frozen inputs, the endpoint, shorter-table, CSE and validation stage takes 885.1 seconds with peak child memory of 3,647 MiB. The added robust-prior and clinical analyses have their own timing records. The application stage takes 33.3 seconds from its prepared inputs. These stages have different starting inputs and are not interchangeable timing estimates.

The original measurements are retained as records of those workflows. The current R/Rcpp replay records its own phase times, software versions and hardware. It recomputes operating characteristics and verifies the frozen rules and gain bounds. Fresh fitting is a

separate operation with explicit search limits. The saved tables define the article's results.

The package supplies the complete decision tables, planning distributions, evaluation curves, risk and gain inputs, and a mapping from every article table to its calculation. All analysis and plotting programs use R or Rcpp. Final trial use requires only the observed counts and the prespecified table.

References

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